Department of Computer Engineering

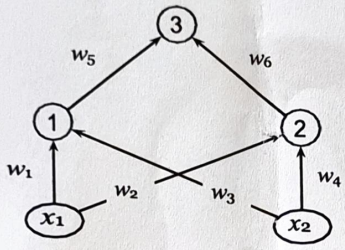
National Institute of Technology Kurukshetra

Mid Term- I, Even Semester 2024-2025

Subject: Deep Learning (CSTC 308/ITTC 308) **Branch:** B.Tech. CS and IT (VI Semester)

Maximum Marks: 15 **Time:** 50 Minutes **Date:** 21/02/2025 **Roll No:** 12212129

1. Consider the Feed-forward neural network given below.



Assume that all internal nodes and output nodes compute the sigmoid function (g). In this question, you will derive an explicit expression that shows how back propagation (applied to minimize the least squares error function) changes the values of $w1$, $w2$, $w3$, $w4$, $w5$ and $w6$ when the algorithm is given the example ($x1$, $x2$, y) with y being class variable (there are no bias terms). Assume that the learning rate is η. Let $o1$ and $o2$ be the output of the hidden units 1 and 2 respectively. Let $o3$ be the output of the output unit 3.

Hint: Derivative of Sigmoid function $g'(x) = g(x)(1 - g(x))$

(i) Forward propagation. Write equations for $o1$, $o2$ and $o3$. [Marks: 3]

(ii) Backward propagation. Write equations for $\delta 1$, $\delta 2$ and $\delta 3$ where $\delta 1$, $\delta 2$ and $\delta 3$ are the values propagated backwards by the units denoted by 1, 2 and 3 respectively in the neural network. [Marks: 3]

Note: δ is the derivative of the loss with respect to the pre-activation output of neuron ($\frac{\partial L}{\partial a}$), where L is the loss function and a is pre-activation output of the neuron.

(iii) Give an explicit expression for the new (updated) weights $w1$, $w2$, $w3$, $w4$, $w5$ and $w6$ after backward propagation. [Marks: 4]

2. McCulloch Pitts (MP) neuron aggregates the inputs and takes a decision based on this aggregation. Consider the following boolean function:

$$f\,(x1, x2, x3) = (x1\ AND\ x2)\ AND\ (!x1\ AND\ !x3)$$

What should be the value of the threshold (θ) such that the MP neuron implements the above boolean function? [Marks: 3]

3. What are the differences between a Perceptron and a Sigmoid Neuron? [Marks: 2]